Beldex Meet — Prototype

Walkthrough across Phase 1 (calling, now a full meeting experience), Phase 2 (identity sign-in) and Phase 3 (payments, now started), with the current flows and fresh web + mobile screenshots.

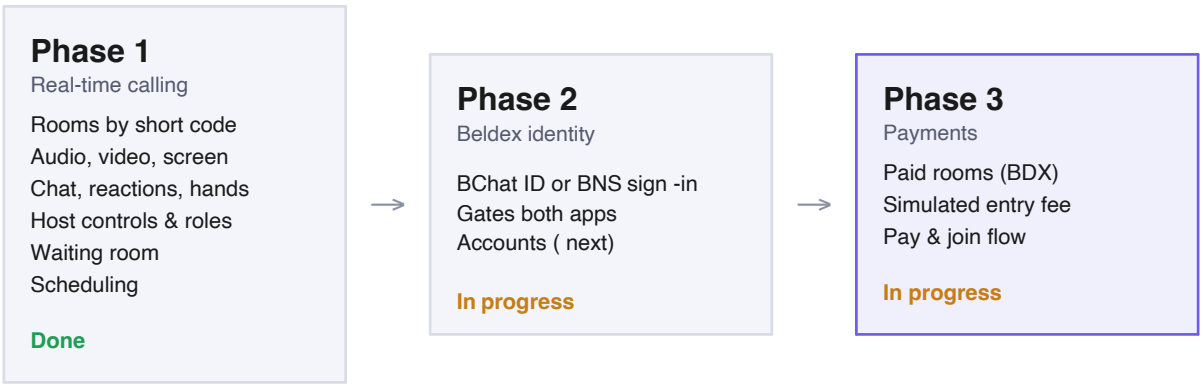
Updated 2026-11-07

Overview

Beldex Meet is a privacy-focused video-meetings app with a browser web client and a native mobile app (iOS + Android), both backed by one token / host-control service and one LiveKit media server. Create or join a room by a short code and talk over camera, microphone, screen share and chat — now with the full in-meeting control set you would expect from a mature conferencing app.

The product is planned in three phases. **Phase 1 (calling) is done** and has grown from basic calls into a complete meeting experience: waiting room, host controls, roles, private chat, reactions, scheduling and in-call audio/video settings. **Phase 2 (identity)** gates both apps behind a BChat ID or BNS sign-in (the wallet signature is still simulated). **Phase 3 (payments) has now started** a room whose code begins with 9 charges a simulated BDX entry fee that the joiner confirms before entering.

Phase	Scope	Status
Phase 1	Real-time calling across web + mobile, plus the full meeting control set (waiting room, host tools, private chat, scheduling, in-call settings)	Done
Phase 2	Beldex identity — BChat ID or BNS sign-in (simulated) that gates both apps	In progress
Phase 3	Payments — paid rooms with a simulated BDX entry fee	In progress



Roadmap — Phase 1 done, Phase 2 and Phase 3 both in progress

What's new since the last prototype

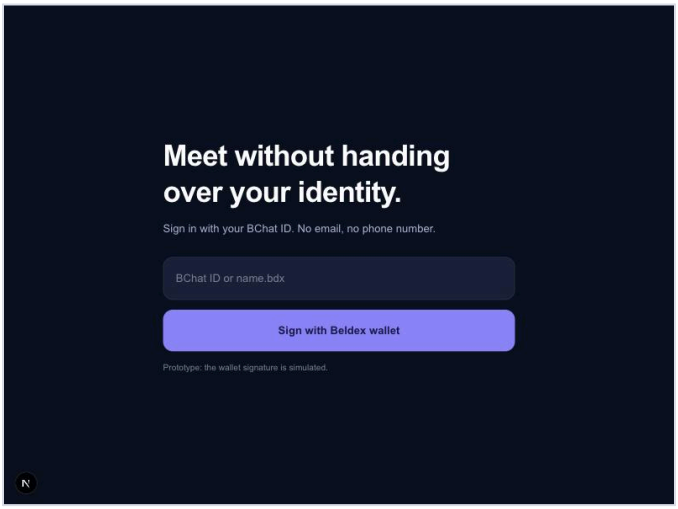
The previous walkthrough showed basic calls with mic, camera, screen share and public chat. Since then the meeting experience has been built out to full feature parity with mainstream conferencing tools, on **both** web and mobile, and the first payments feature has landed.

- **Host controls:** mute all / mute one, ask to unmute, stop a participant's video, rename, remove, lower all hands, lock meeting, end meeting for all.
- **Roles:** host, co-host and make-host (transfer) — now token-authenticated, so the host can no longer be spoofed.
- **Waiting room:** admit / deny / admit all, with join and waiting toasts.
- **Server-enforced permissions:** "Allow participants to" unmute, start video, share screen, chat & react, rename.
- **Private 1:1 chat** alongside public chat, with timestamps and an unread badge.
- **Reactions & raise hand**, floating self-view (hide / mirror), and speaker-vs-gallery view.
- **In-call settings:** pick microphone / speaker / camera, input-level meter, speaker test tone, noise suppression and echo cancellation.
- **Meeting info & invite:** code + live duration timer, invite by link / invitation / email / .ics from inside the call.
- **Scheduling:** create, list, start, invite, add-to-calendar and delete upcoming meetings.
- **Paid rooms (Phase 3):** "Charge entry fee" on create and a "Pay N BDX & join" confirmation when joining.

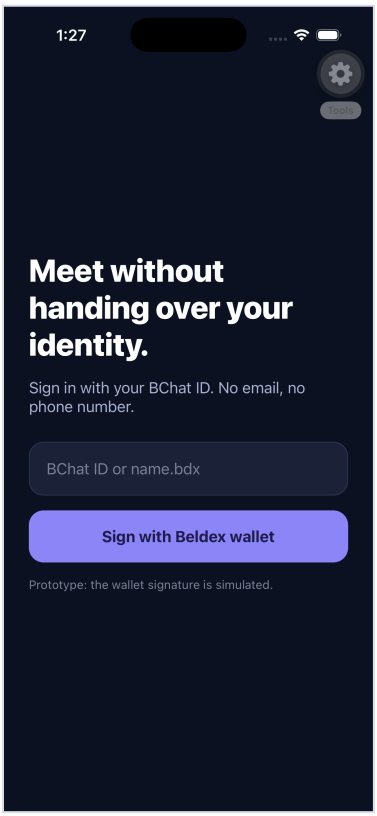
Sign in — Phase 2 (identity)

Both apps open to a Beldex sign-in that gates the rest of the app. You enter a BChat ID or a *name.bdx* handle and tap **Sign with Beldex wallet** — no email, no phone number. For the prototype the wallet signature is simulated; the signed-in name carries into meetings, and you can sign out from the home screen.

Demo credentials: **demo.bdx**, alice.bdx, bob.bdx, carol.bdx, or a 66-char **05…** BChat ID — any *name.bdx* or valid 05… ID also works.



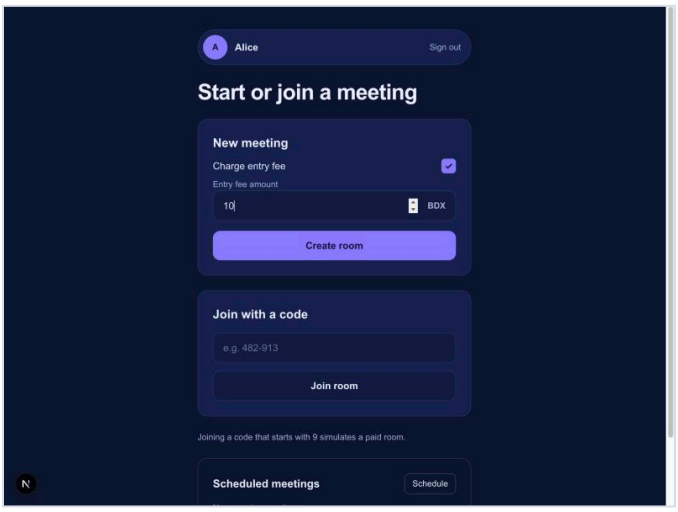
Web sign-in



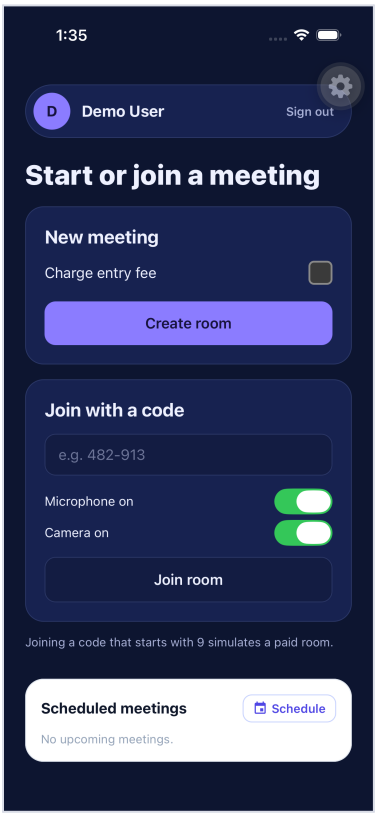
Mobile sign-in

Home — start or join a meeting

After signing in, the home screen shows the signed-in identity with a sign-out, a **New meeting** card and a **Join with a code** card. New in this build: a **Charge entry fee** toggle (Phase 3) that creates a paid room and stores the fee in BDX, mic/camera join preferences on mobile, and a **Scheduled meetings** list. A code that starts with 9 is a (simulated) paid room.



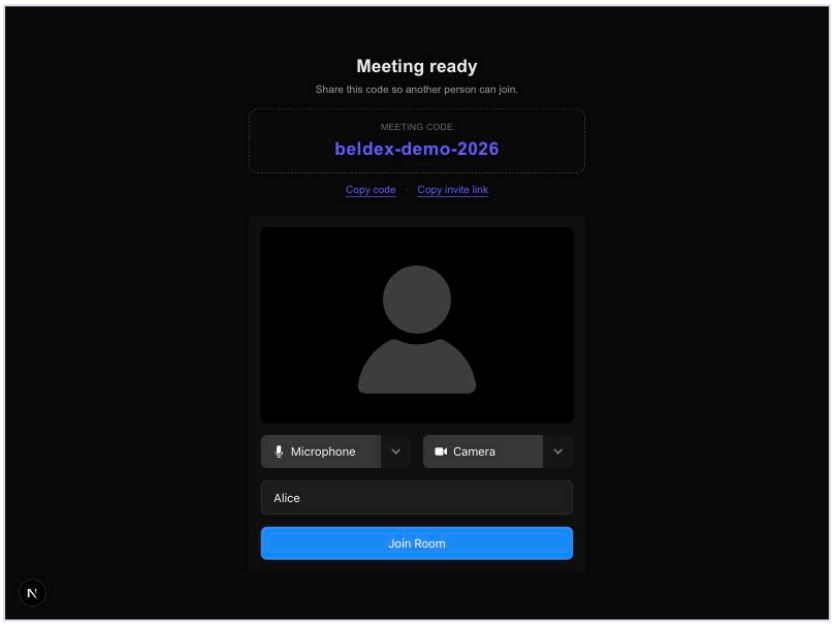
Web home — “Charge entry fee” set to 10 BDX



Mobile home — fee toggle, mic/camera, scheduling

Joining a call

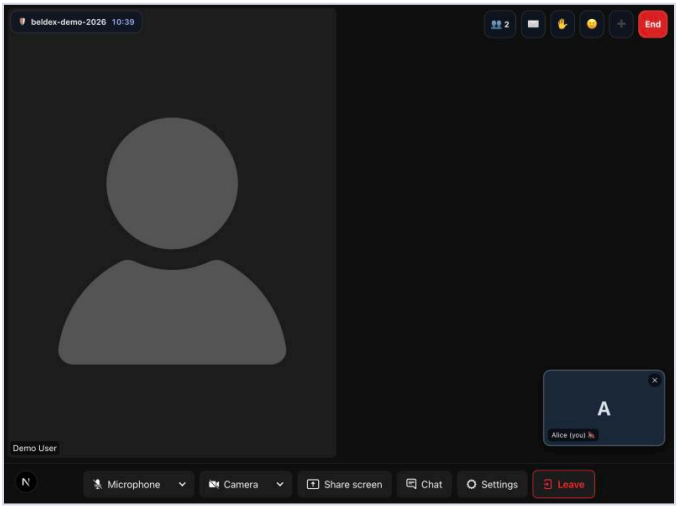
On the web, a room opens on a **Meeting ready** pre-join screen that shows the shareable code, a camera/mic picker and a name field (prefilled from your identity) before you join. The mobile app offers the same with Microphone / Camera switches on the home card. Below, the web host is signed in as **Alice** in room *beldex-demo-2026*.



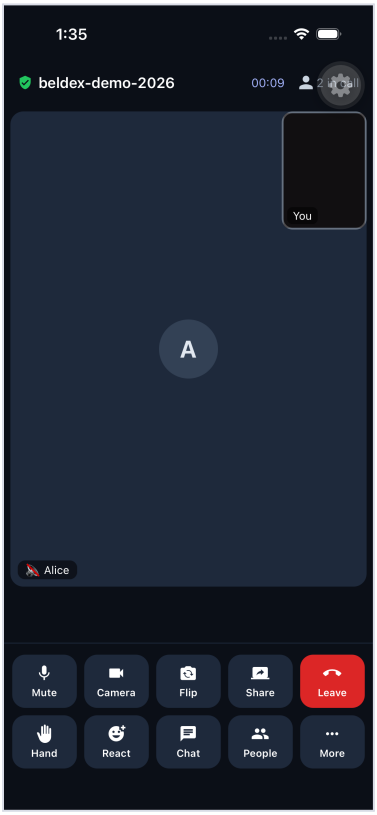
Web pre-join — shareable code, device pickers and name before joining

In the meeting

Once in the call, the grid shows a tile per participant with a control bar. The header carries a security/lock shield, the room code and a live duration timer; a floating, draggable self-view sits in the corner. Below is the **same live call** seen from both clients: the web host (Alice) and the phone (Demo User) are in room *beldex-demo-2026* together — “2 in call”.

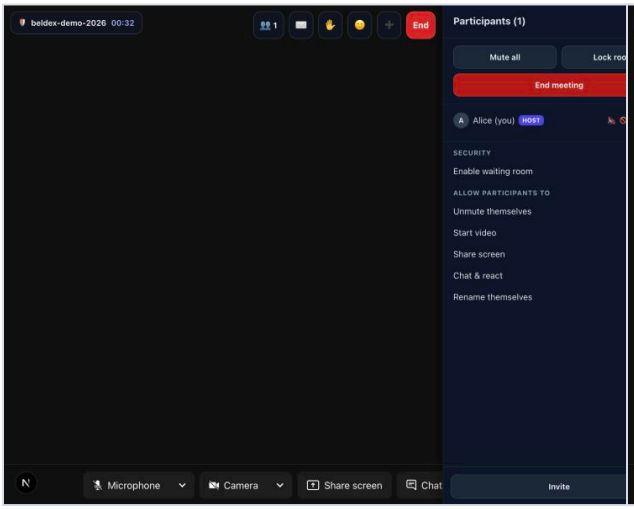


Web in-call — Demo User's tile, timer, control bar, End

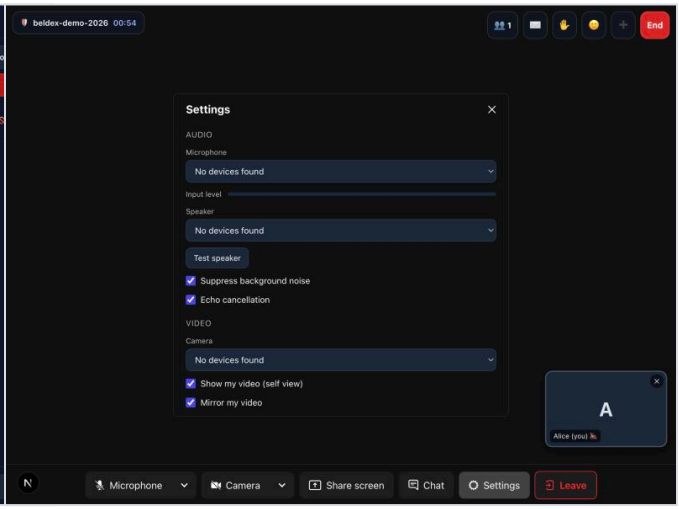


Mobile in-call — Alice's tile, self-view, full controls

The control bar carries Microphone, Camera, Share screen, Chat, Settings and Leave, with Participants, Private messages, Raise hand, React, Invite and (for the host) End across the top. The **Participants** panel is where the host manages the room; the **Settings** panel holds the device and audio-processing options.

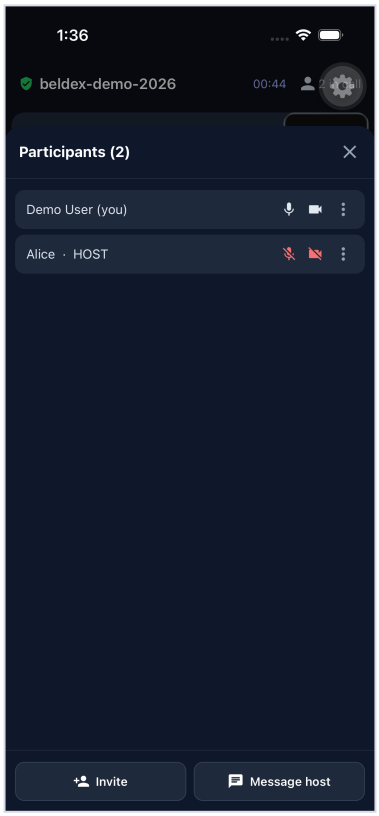


Participants — roles, host tools, waiting room and permission toggles



Settings — devices, test tone, noise suppression, self-view

On mobile the same management lives in a **People** sheet: the participant list with roles (HOST), live mic/camera state, a per-person actions menu, plus Invite and a private "Message host".



Mobile People sheet — roles, mic/camera state, per-person actions

Paid rooms — Phase 3 (payments, started)

Phase 3 adds the Beldex payment layer on top of Phase 1 calling and Phase 2 identity. The first slice is live in both apps: when you create a meeting you can turn on **Charge entry fee** and set an amount in BDX. That produces a room code beginning with 9 and stores the fee on the server so anyone joining sees it. Joining such a code shows a **Pay N BDX & join** confirmation (shown on the web home screenshot above). Anticipated next:

- Real Beldex payments tied to a user's identity (the transfer is simulated today).
- Monetized meetings — paid rooms, paid time, or premium features.
- Billing / entitlement checks gating room creation or specific capabilities.

Like Phase 2, this builds on the existing stack rather than replacing it — payments and entitlements gate access; the calling experience is unchanged.

Architecture & tech stack

Both clients share one backend. A client fetches a short-lived token from the Next.js server (GET /api/token?room&username), which returns a room-scoped LiveKit JWT plus the serverUrl; the client then connects directly to LiveKit, which relays audio/video/screen among participants. Host actions (mute, remove, lock, admit, roles, end…) go through /api/room/* and are authenticated with the caller's own join token, verified server-side — a forged request is rejected. Room settings live in LiveKit room metadata; scheduled meetings and room fees are stored as JSON on the server. Sign-in, the wallet signature and

payments are still client-side and simulated.

Piece	What it is	Where
LiveKit media server	WebRTC SFU for A/V + screen	port 7880
Token + host-control server	Next.js 15; mints JWTs at /api/token, host & meeting APIs at /api/room/* and /api/meetings	port 3003
Web client	Next.js / React, LiveKit components	browser
Mobile client	Expo / React Native, @livekit/react-native	iOS + Android

Secrets live in .env.local (dev defaults devkey / secret). The mobile app needs a native dev build (LiveKit WebRTC modules), so it cannot run in Expo Go. For reliable media to a physical device, LiveKit Cloud is recommended over a local dev server.

Status & what's still simulated or pending

- The wallet signature (Phase 2) and payments (Phase 3) are simulated — there is no real signature or on-chain transfer yet.
- iOS shares the app's own screen only; full-device screen share needs a Broadcast Upload Extension. Android shares the whole device screen.
- Scheduled meetings and room fees are JSON files on the Next.js server (fine for one instance; use a database for production).
- In-meeting chat history is in-memory for the duration of the meeting.
- Not yet in scope: cloud recording, live captions / transcription, breakout rooms, polls, whiteboard and virtual backgrounds.